



IIT ROORKEE



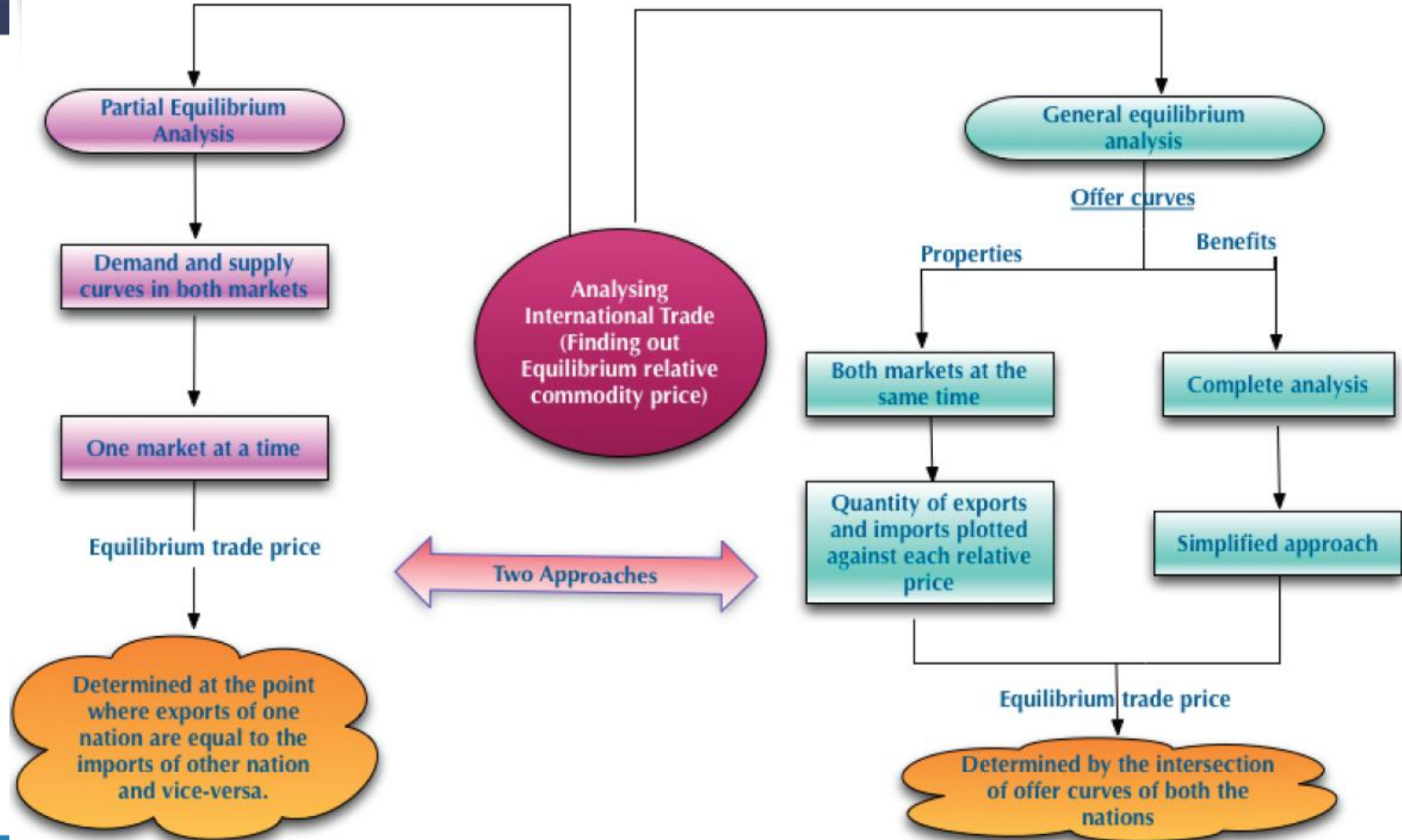
NPTEL ONLINE
CERTIFICATION COURSE

OFFER CURVES IN TRADE ANALYSIS

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Introduction

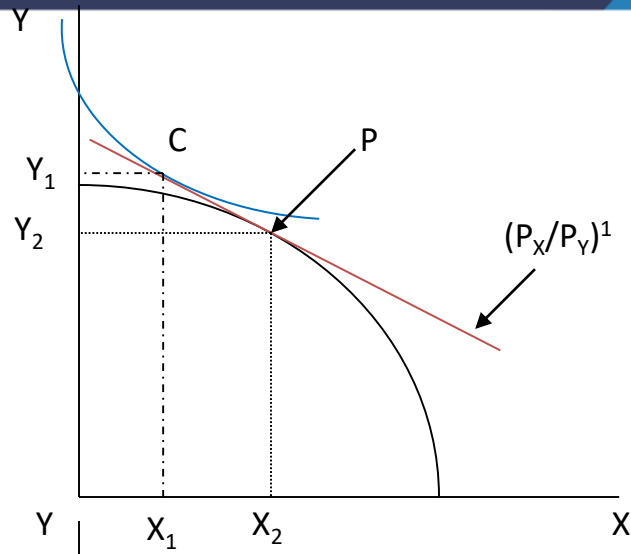
- Relative commodity price differences between two nations in isolation reflect comparative advantage, and forms basis for mutually beneficial trade.
- Can use partial and general equilibrium analysis to determine equilibrium-relative commodity price at which trade will take place.
- **SMART**, the market access simulation package included in WITS, is a partial equilibrium modelling tool.
- **GTAP** for general equilibrium



Offer Curves

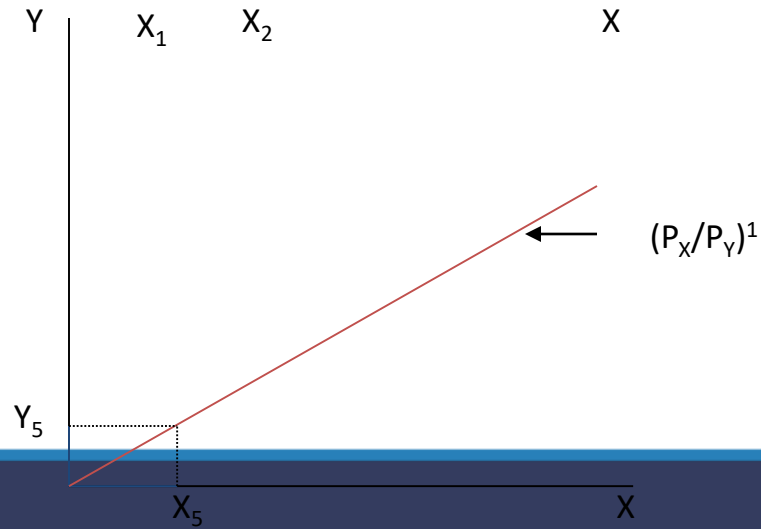
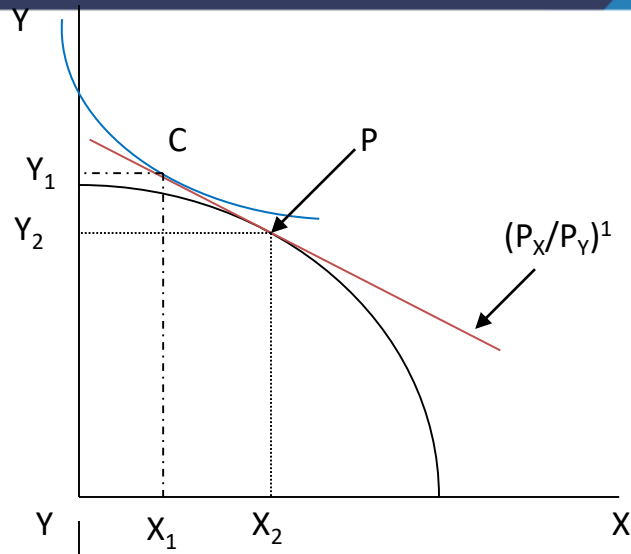
- **Offer curves** introduced to international economics by *Marshall* and *Edgeworth*.
- Shows how much of its import commodity a nation demands for it to be willing to supply various amounts of its export commodity.
- Can be derived from production possibilities frontier, indifference map and various hypothetical relative commodity prices at which trade could take place.
- all combinations of a country's desired exports and imports at different terms of trade
- also known as **reciprocal demand curves (J.S. Mills)**
- measures of willingness to trade

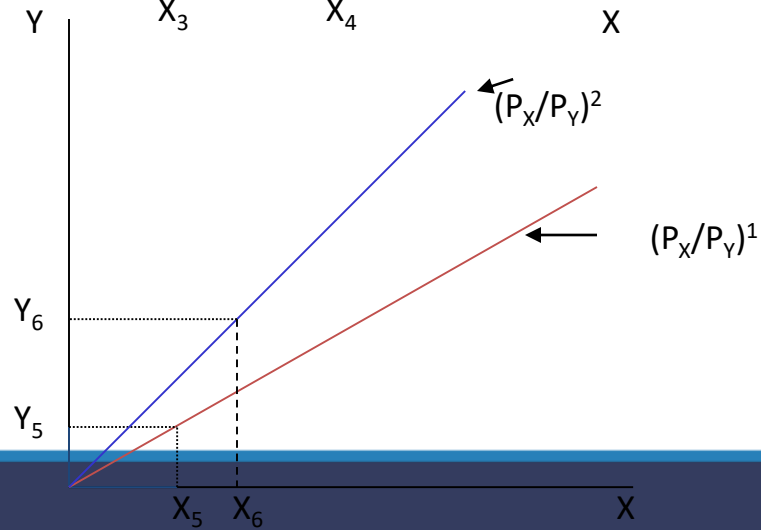
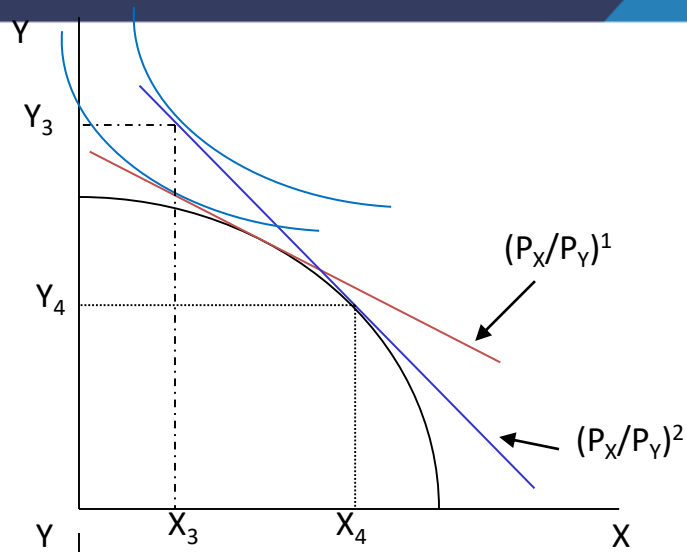


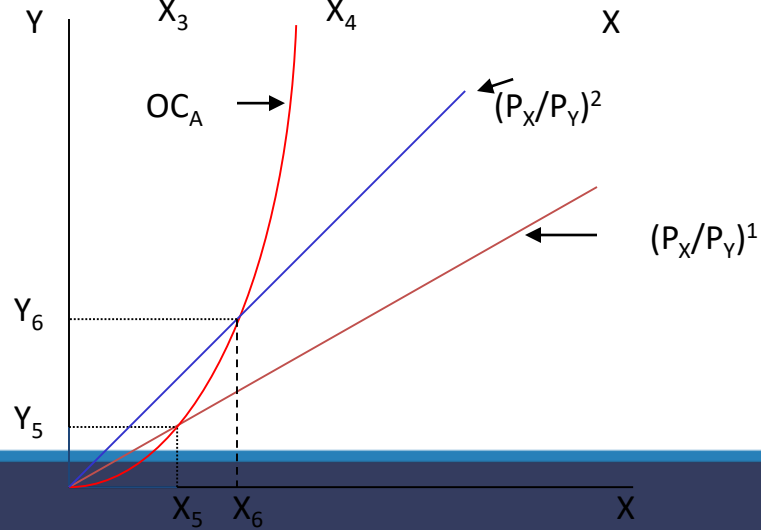
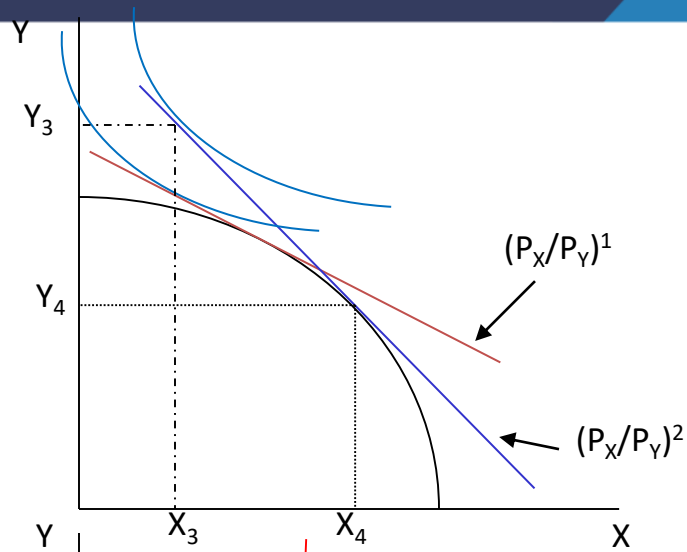


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Offer Curves

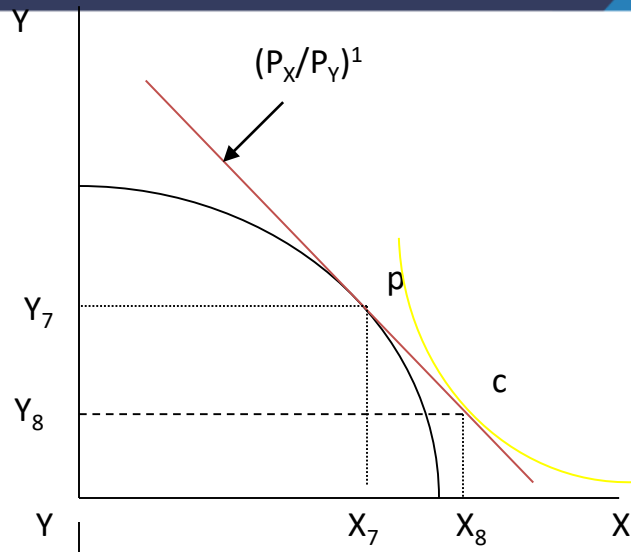
- Offer curves represent willingness to trade at every possible terms of trade
- As the relative price of good X rises, Country A becomes willing to export more and import more
- Offer curves “bow” towards the import good axis



Deriving Country B's Offer Curve

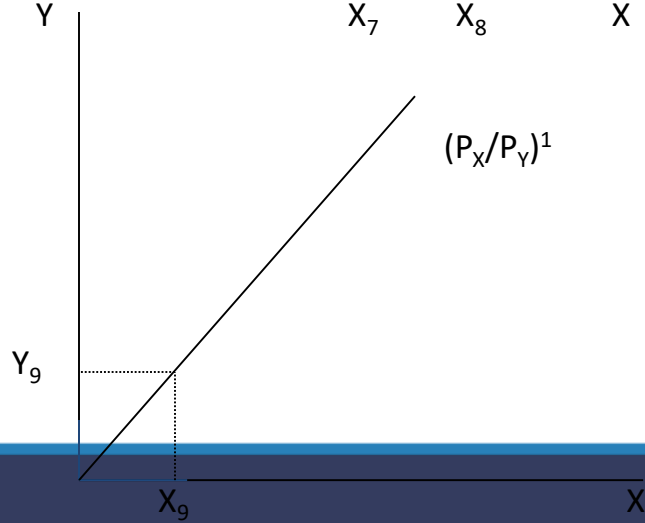
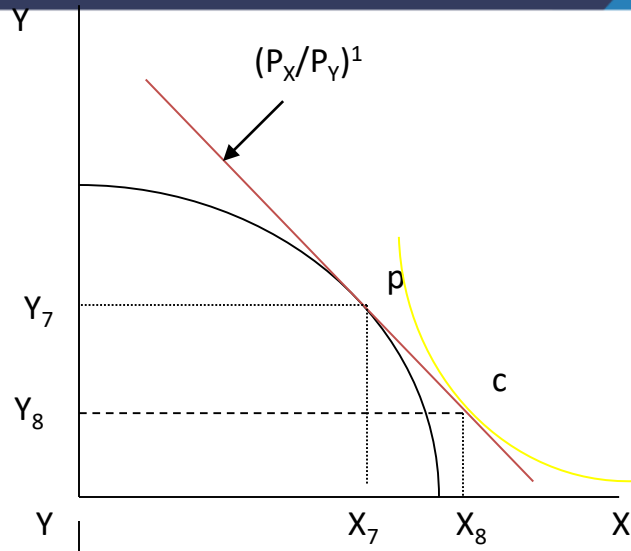
- This will reflect Country B's willingness to trade at different terms of trade
- B's offer curve bows towards the axis with B's import good on it

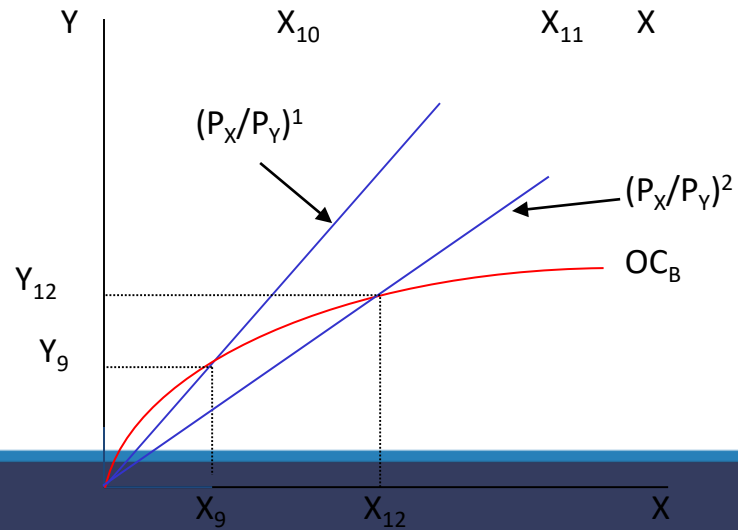
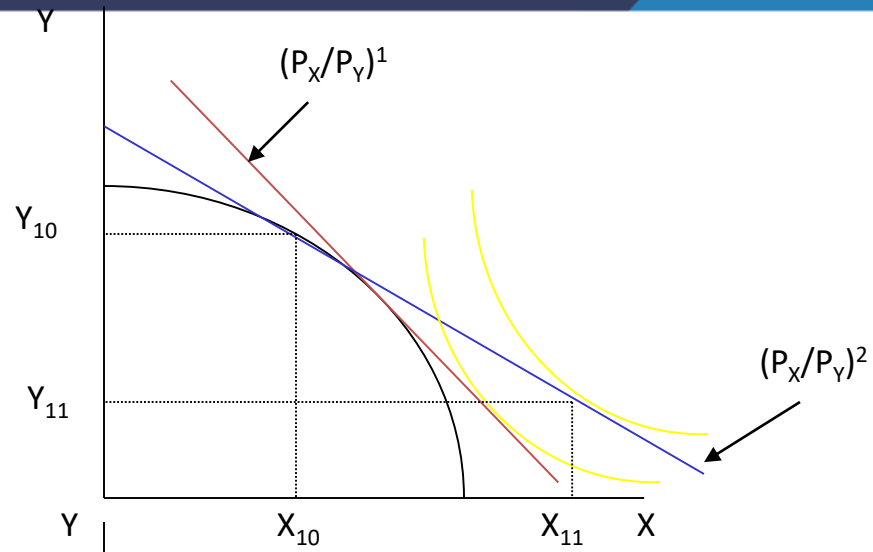




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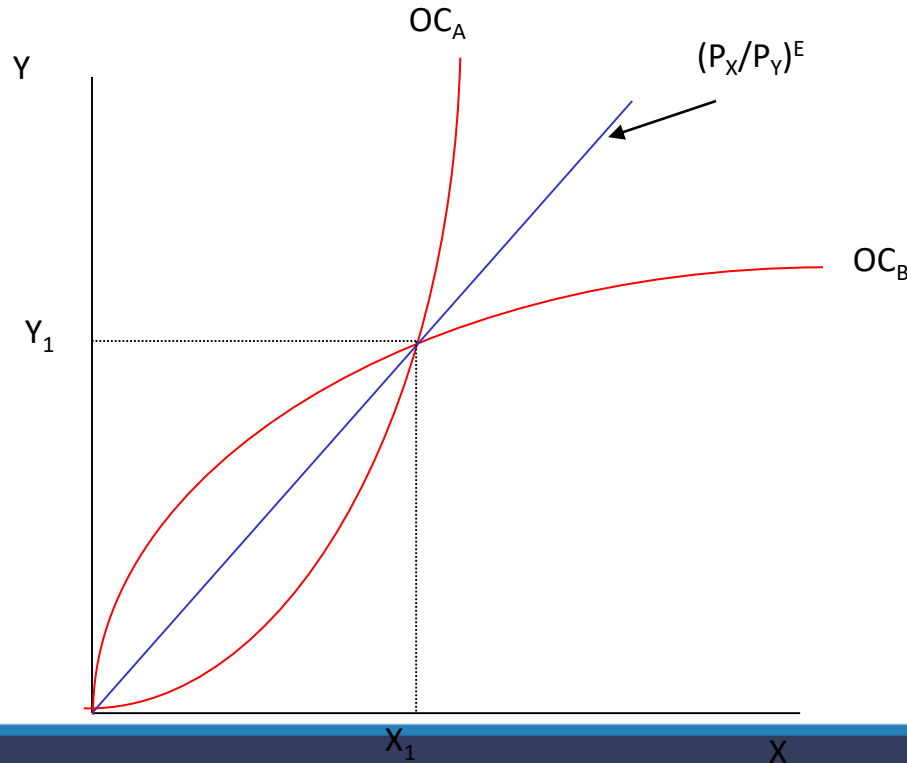


Terms of Trade Equilibrium

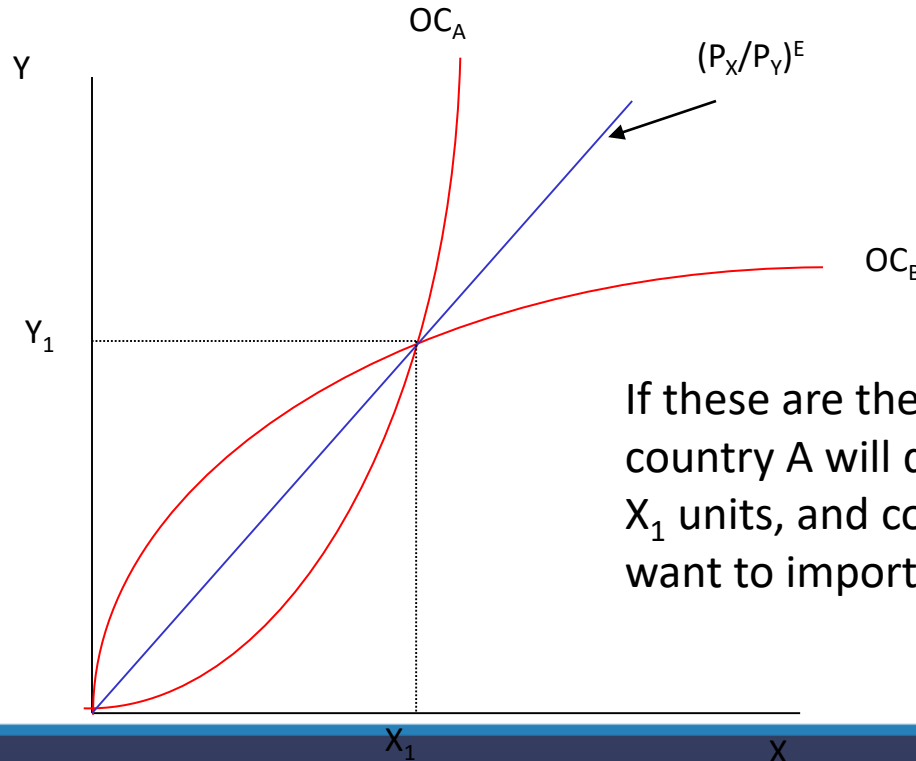
- The international terms of trade (that is, P_X/P_Y) will be the slope of a line passing through the point where the offer curves cross.
- This equilibrium point takes into account demand and supply conditions in both countries



Terms of Trade Equilibrium

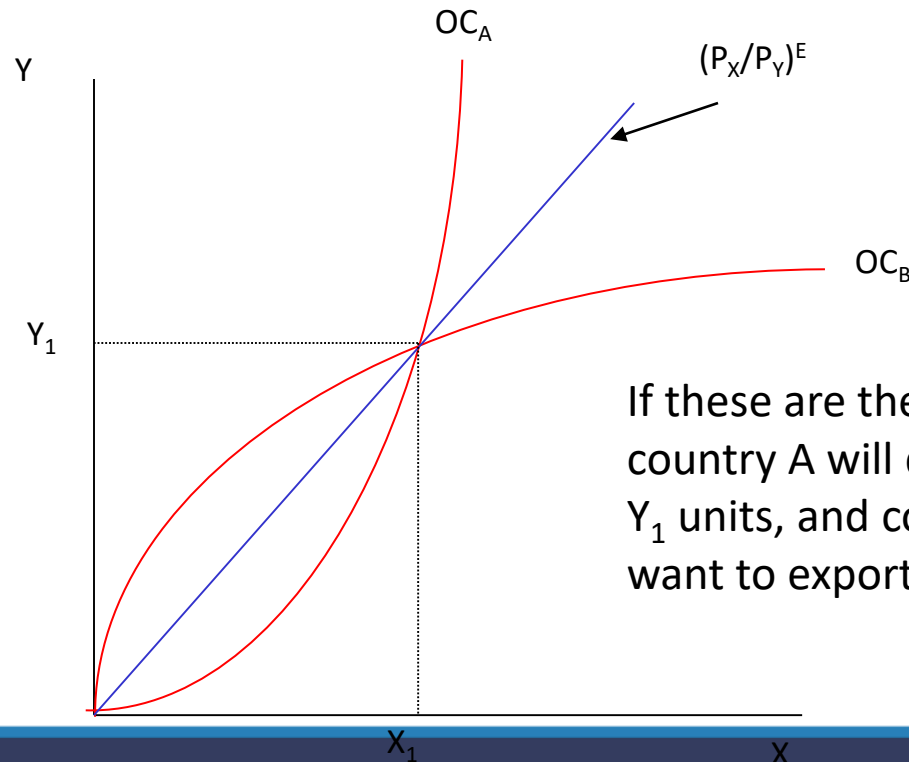


Terms of Trade Equilibrium



If these are the terms of trade, country A will desire to export X_1 units, and country B will want to import X_1 units

Terms of Trade Equilibrium



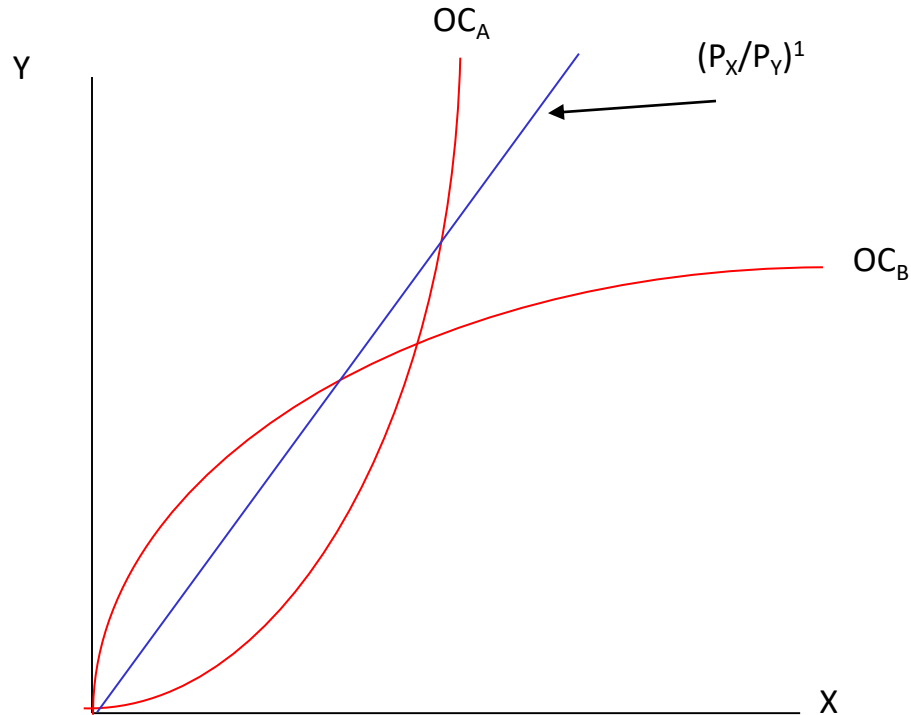
If these are the terms of trade, country A will desire to import Y_1 units, and country B will want to export Y_1 units

Is It the Equilibrium?

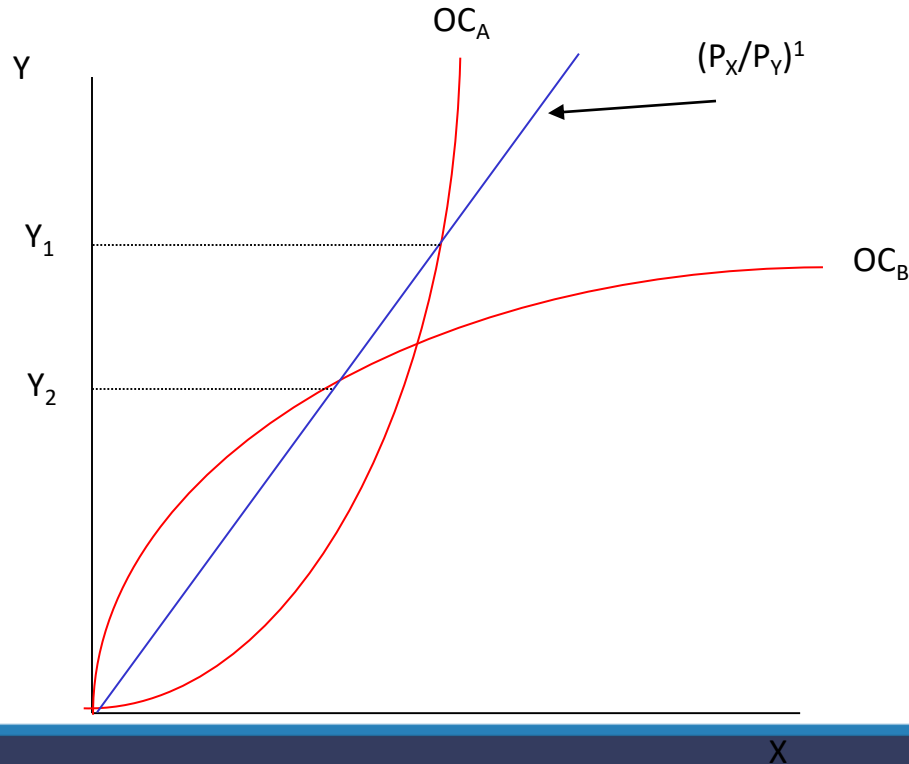
- Any terms of trade other than $(P_X/P_Y)^E$ will result in
 - excess demand for one good
 - excess supply for the other
- Therefore relative prices will adjust until $(P_X/P_Y)^E$ is reached



Disequilibrium

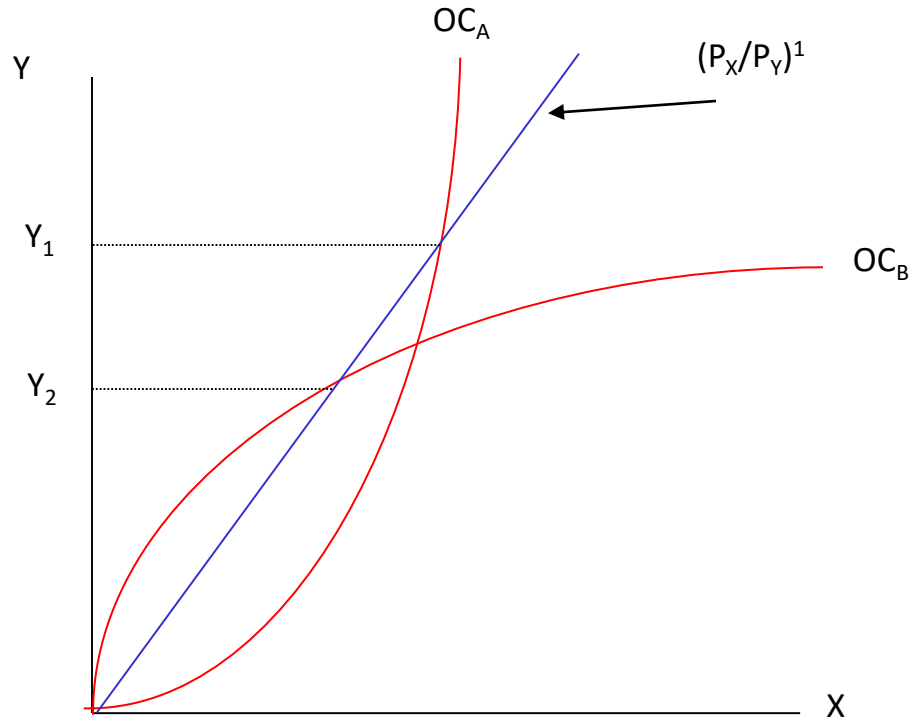


Disequilibrium

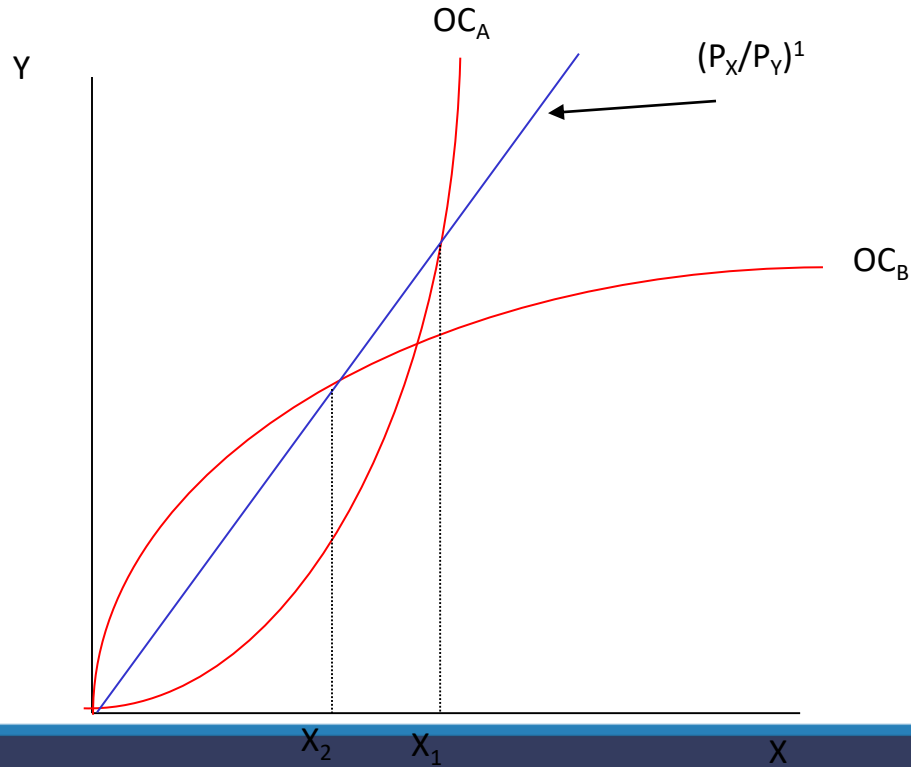


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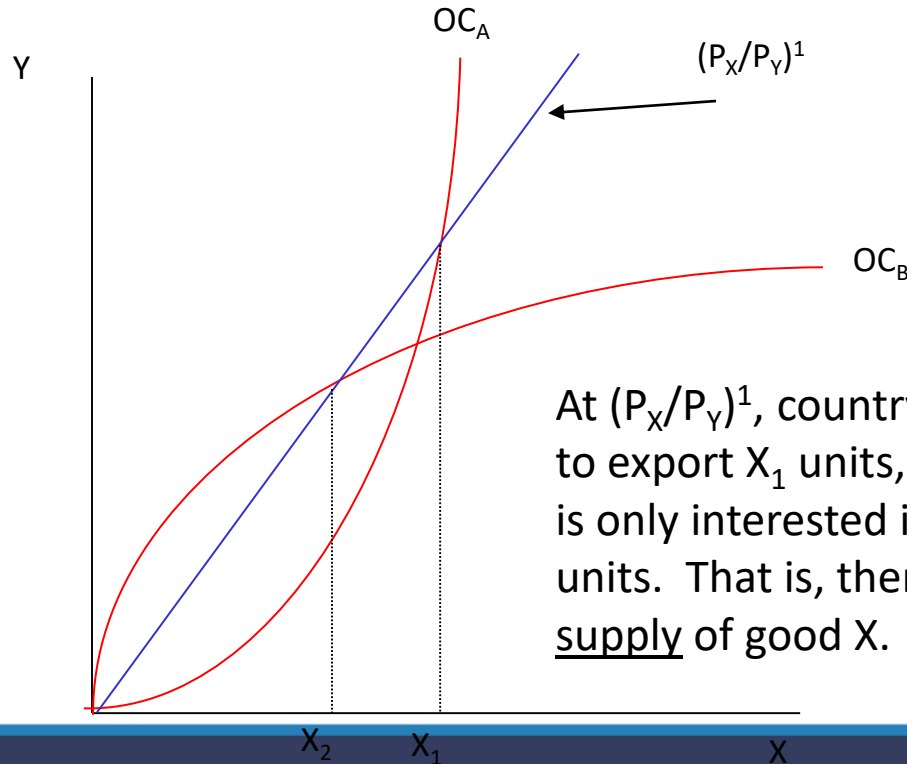
Disequilibrium



Disequilibrium



Disequilibrium



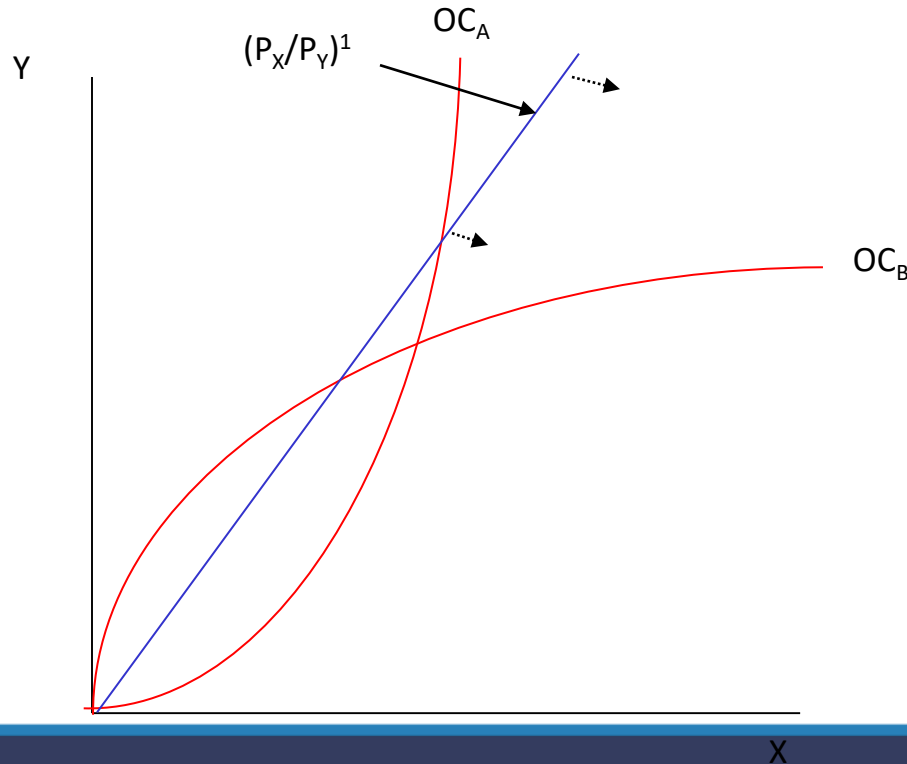
At $(P_X/P_Y)^1$, country A wishes to export X_1 units, but country B is only interested in importing X_2 units. That is, there is an excess supply of good X.

Disequilibrium

- Excess demand for Y causes P_Y to rise
- Excess supply of X causes P_X to fall
- Thus, (P_X/P_Y) falls
- In other words, the terms of trade line gets flatter, moving the countries in the direction of equilibrium



Moving Towards Equilibrium



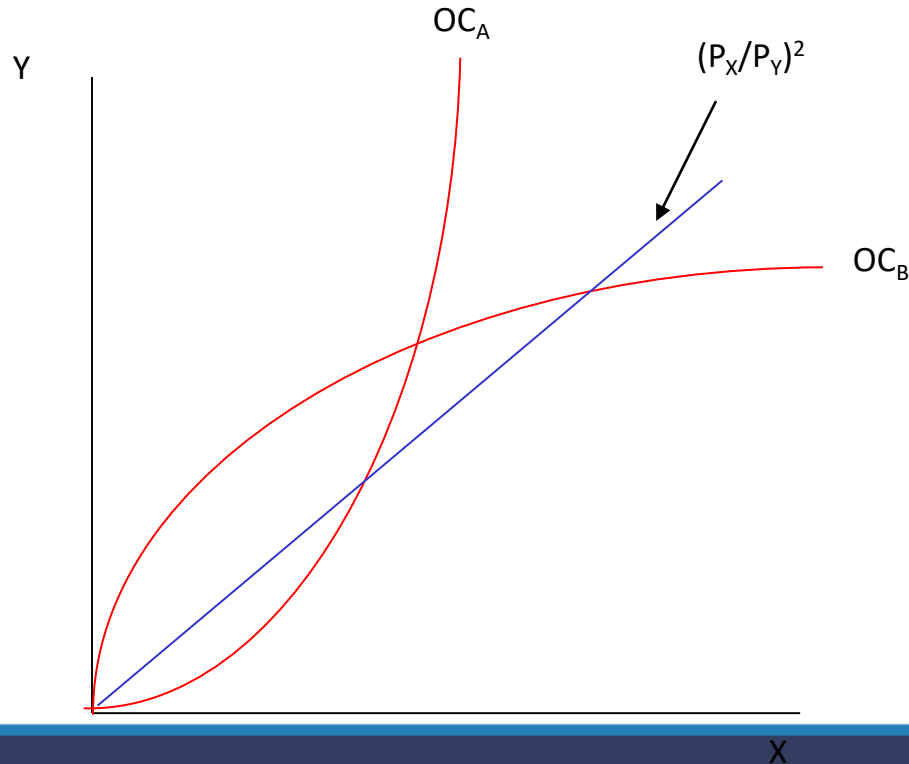
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Disequilibrium

- Terms of trade lines that are flatter than $(P_X/P_Y)^E$, such as



Disequilibrium



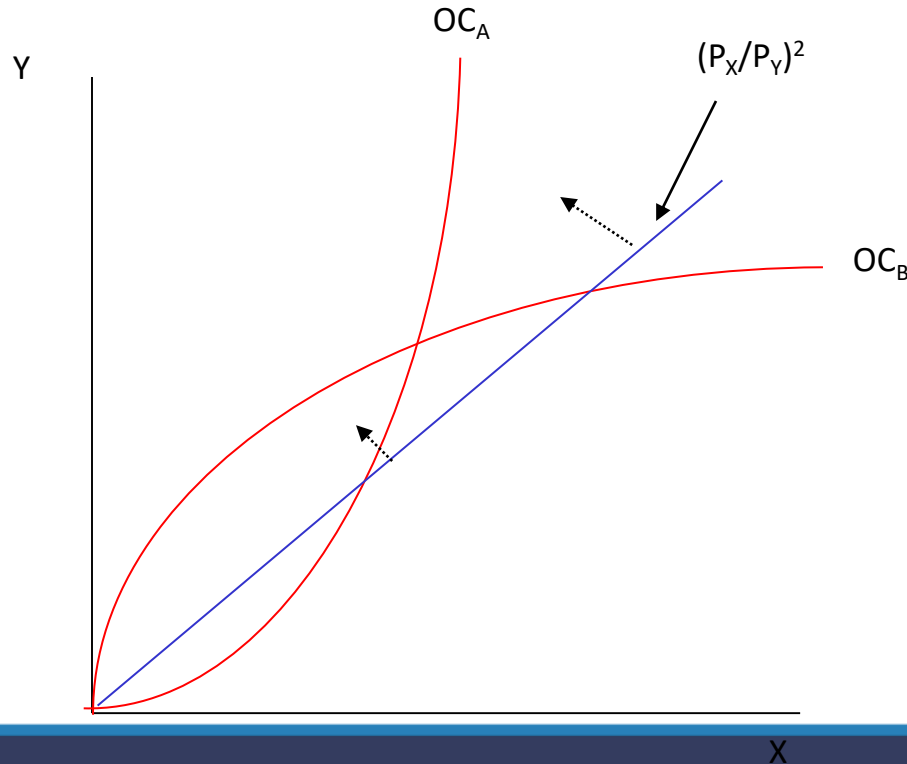
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Disequilibrium

- Terms of trade lines that are flatter than $(P_X/P_Y)^E$ will result in
 - an excess demand for good X
 - an excess supply of good Y, and so
- (P_X/P_Y) will rise
- That is, the terms of trade line will get steeper until $(P_X/P_Y)^E$ is reached



Moving Towards Equilibrium



X

Shot Note on Terms of Trade

- A country's "terms of trade" are the price of its exports divided by the price of its imports, so a rising terms of trade is good news
- In this example, (P_X/P_Y) is country A's terms of trade, since A exports good X and imports Y
- (P_Y/P_X) is country B's terms of trade in this example



Short Note on the Terms of Trade, continued

- As A's terms of trade (P_X/P_Y) improve, B's terms of trade (P_Y/P_X) must be deteriorating and vice-versa

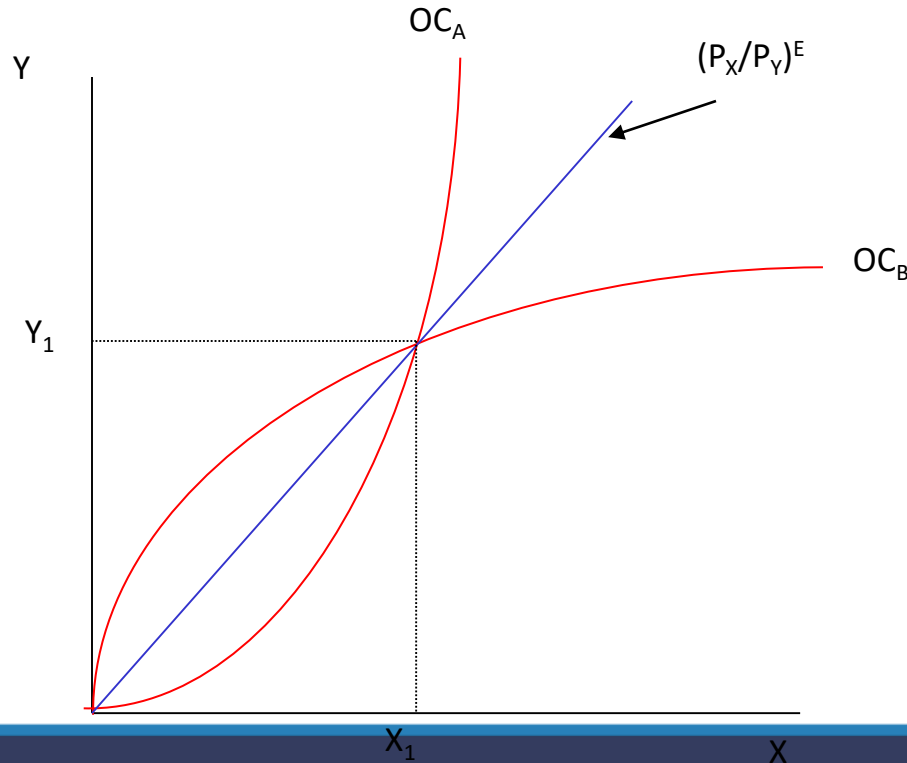


Shifts of Offer Curves

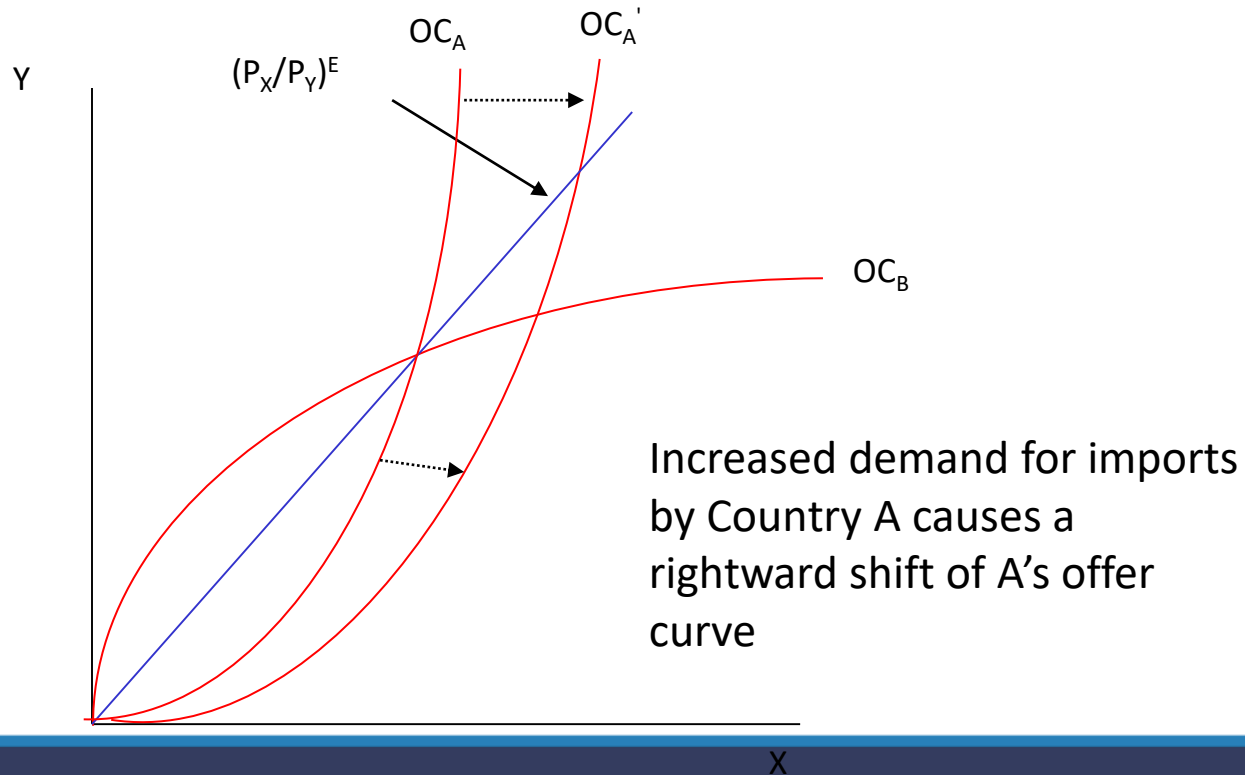
- Anything that causes country A's willingness to trade to change will shift A's offer curve
 - increased willingness to trade: OC_A shifts right
 - decreased willingness to trade: OC_A shifts left
- These can be caused by
 - changes in demand conditions or
 - changes in supply conditions



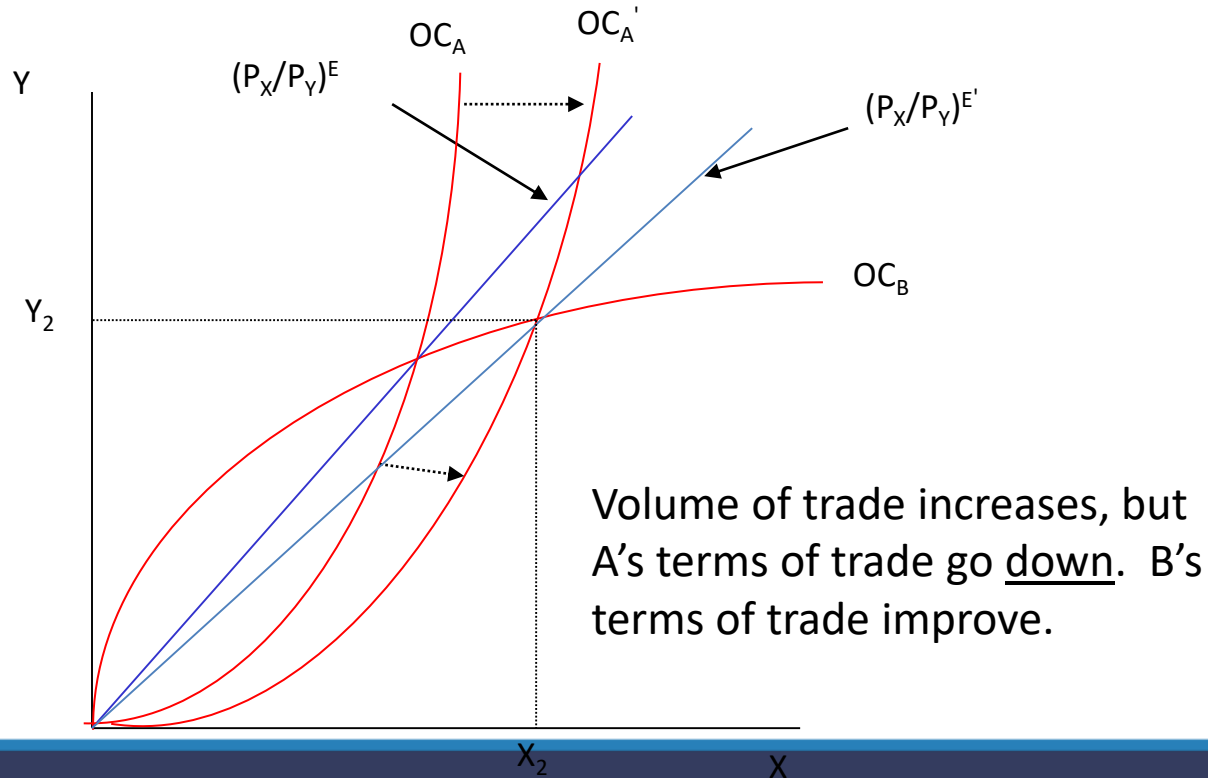
Demand Changes in Country A



Demand Changes in Country A



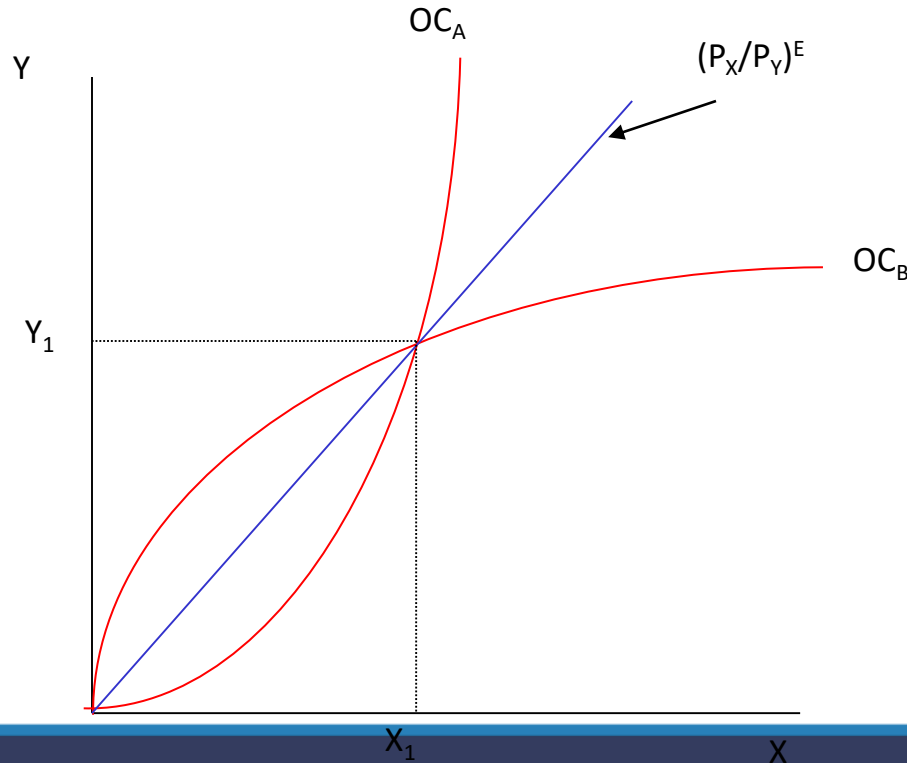
Demand Changes in Country A



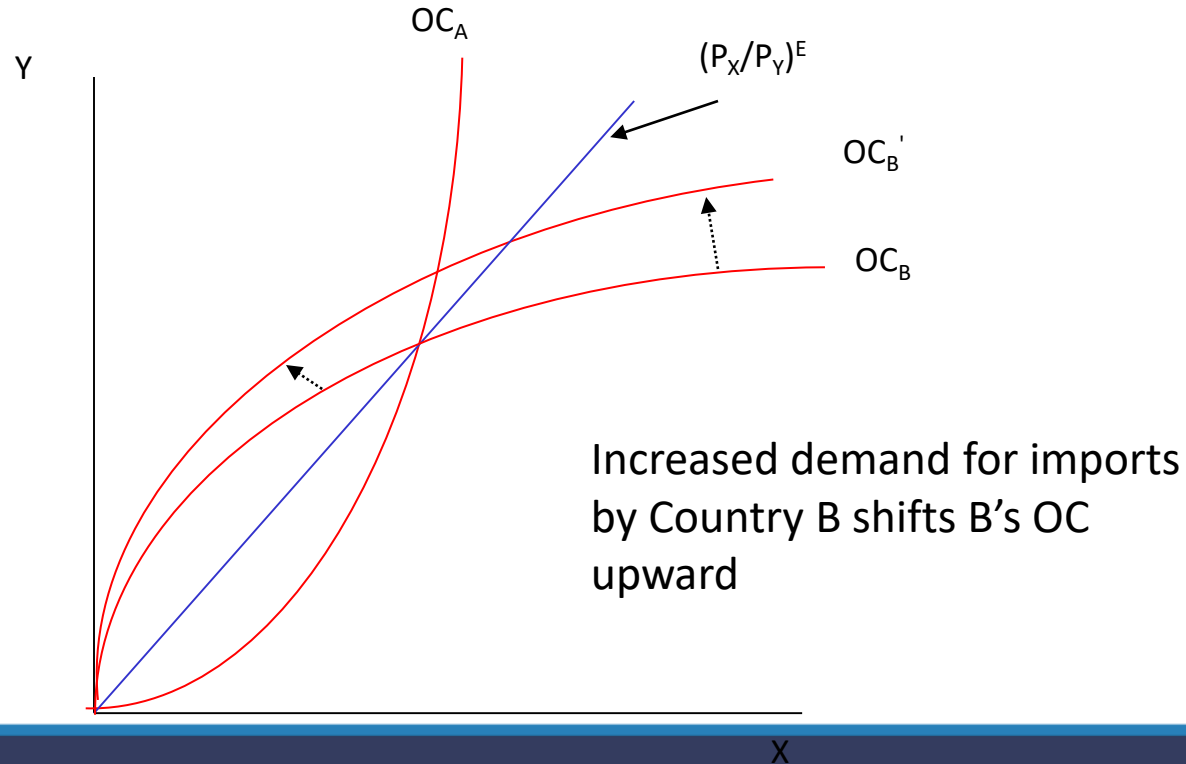
Demand Changes in A

- Any change that might make A demand more imports leads to a rightward OC shift, and thus
 - an increase in trade volume
 - a decrease in A's terms of trade

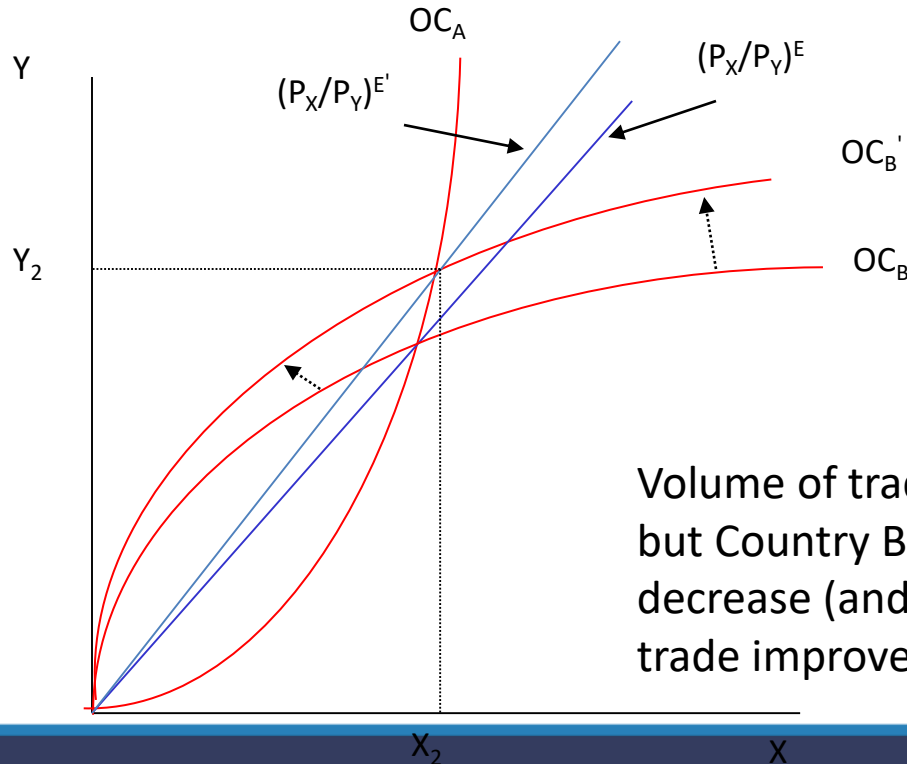
Demand Changes in Country B



Demand Changes in Country B



Demand Changes in Country B



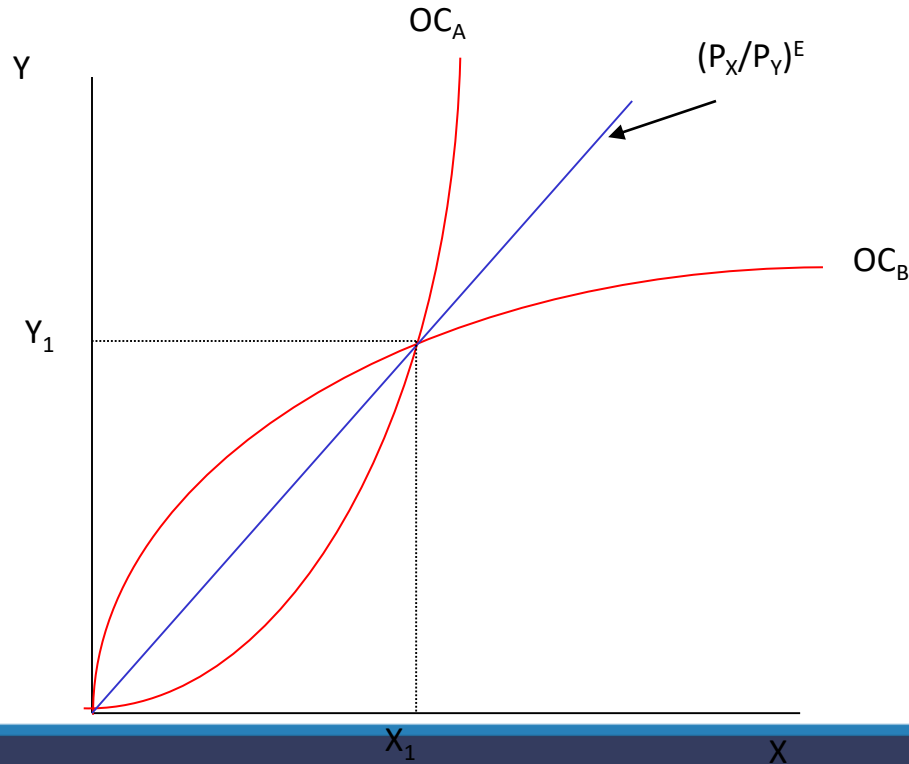
Volume of trade increases,
but Country B's terms of trade
decrease (and A's terms of
trade improve).

Other Demand Changes

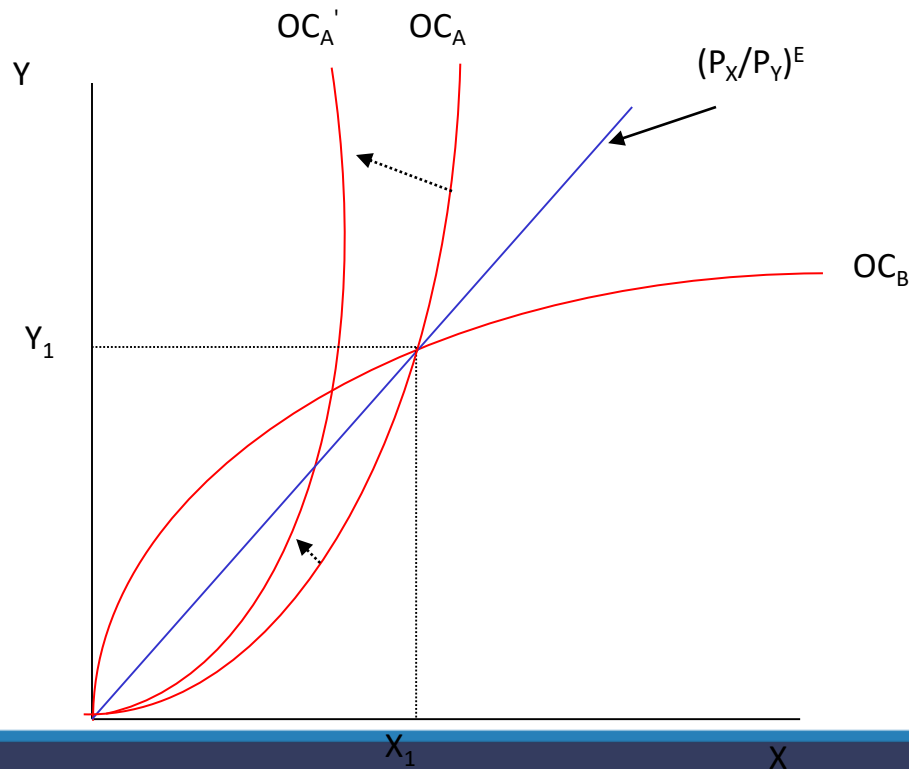
- Any decrease in a country's willingness to trade will shift its OC leftward or downward
- An example is when a country imposes an import tariff
- Tariffs therefore lead to decreased trade volume, but improve the imposing country's terms of trade



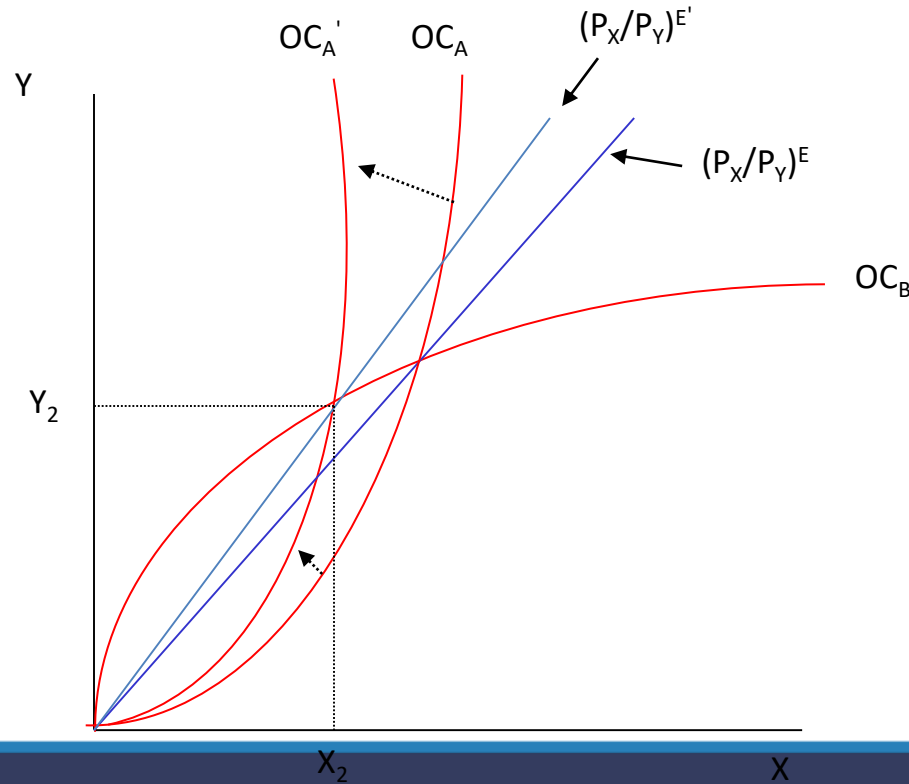
Imposition of Tariff by Country A



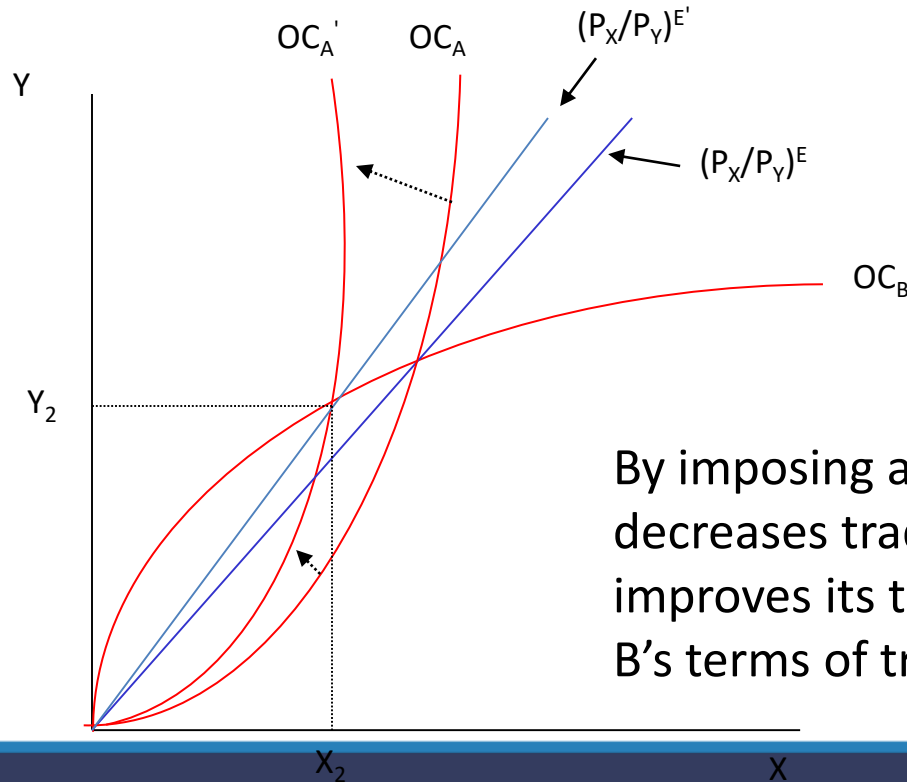
Imposition of Tariff by Country A



Imposition of Tariff by Country A



Imposition of Tariff by Country A



By imposing a tariff, Country A decreases trade volume, and improves its terms of trade (but B's terms of trade deteriorate)

Supply Changes

- Changes in supply conditions will also shift a country's offer curves around
- Examples include
 - productivity changes
 - discovery of new resources



*Thank
you*

